

Explicit Normative Reasoning and Machine Ethics*

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Abstract

The proposed Explicit Normative Reasoning and Machine Ethics (ENoRME) project will contribute foundational technology to foster and enable the development of ethical and legal governors for intelligent autonomous systems. These governors will provide symbolic, deductive means of control that operate orthogonally to and in addition to mechanisms at the less transparent, less accountable, and less trustworthy level of subsymbolic reasoning. In particular, ENoRME will develop a much needed, powerful and flexible, on-demand universal reasoning workbench, with a particular emphasis on normative reasoning. It will explore and demonstrate the use of the workbench in selected case studies and experiments. ENoRME will contribute essential input to the development of “Trustworthy AI”.

1 Introduction

Intelligent autonomous systems (IASs) are rapidly entering applications in industry, military, finance, governance, administration, healthcare, etc., leading to a transition period with unprecedented dynamics of innovation and change, and with unpredictable outcomes [8]. Legislatures, regulatory bodies, intergovernmental organizations, etc., indeed society as a whole, are challenged not only with keeping pace with these potentially disruptive developments, but also with staying ahead and wisely guiding the transition. Since there is much at stake, preemptive investments in the exploration, development, and implementation of appropriate means of controlling IASs are absolutely justified. A balanced approach to AI is needed, fostering positive impacts while preventing negative side effects. This vision is shared in particular by the European Commissions (EC) High-Level Expert Group (HLEG) on Artificial Intelligence (AI) in their recently published “Ethics Guidelines for Trustworthy AI” [13].

Many AI researchers, developers, and users are concerned with challenges such as “How to stay ahead ...”, “How to keep up with ...”, and “How to close the gap to ...” the rapid developments in AI. A global technology race is the result. A much smaller community is concerned with researching and developing means of wisely regulating and controlling future IASs. The proposed **Explicit Normative Reasoning and Machine Ethics** (ENoRME) project addresses these issues. The primary focus is on risk prevention, more precisely, on

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explicit ethical governance of IASs [1]. We see it as a societal mandate to invest in risk prevention technology; such a sensitive issue should not be driven by commercial interests alone. Despite of this focus on “Trustworthy AI”, the range of AI applications that will be enabled by ENoRME is wide and colourful.

2 The State of the Art

The questions of how transparency, explainability, and verifiability can best be achieved in future IASs, and whether bottom-up or top-down architectures should be preferred, are discussed in various papers, e.g., [6, 15, 7, 18]. Dennis et al. [6] make a compelling case for the use of formal verification – a well-established technique for proving correctness of computer systems – in the area of machine ethics. ENoRME will develop a reasoning infrastructure for such normative reasoning. Such an infrastructure is much needed, and related work is thin.

Research and concrete results for the automation of reasoning for quantified deontic logics, and their combinations with other non-classical logics (epistemic, temporal, defeasible, probabilistic, etc.) as required in realistic applications of normative reasoning, is sparse. A connection-based ATP covering among others, first-order standard deontic logic, has been developed by Otten [14]. A tableaux-based propositional reasoner is employed in the work of Furbach and Schon [10]. First-order resolution methods for propositional modal logics have been contributed by Schmidt and Hustadt [16]. Steen and Benzmüller developed Leo-III [17], which has been adapted to the automation of more than 120 different quantified (multi-)modal logics [11], and also for an initial range of quantified deontic logics. Benzmüller, Parent, and van der Torre have collaborated on the automation of propositional and quantified deontic logics, on the development of an overall methodology for the automation of normative reasoning, and on the use of those in some small, selected case studies [3, 2].

There are some first-steps in the development of logical architectures for ethical reasoning that are particularly relevant for the planned work in ENoRME. For example, Carnielli and Bueno-Soler are leading experts in paraconsistent reasoning and logic combinations [4, 5]. Slavkovik, Liao, and van der Torre have collaborated on a methodological level to develop an artificial moral agent architecture that uses techniques from normative systems and formal argumentation to reach moral agreements among stakeholders [12]. Dennis and Fisher have studied ethically critical machine reasoning, and have proposed practical architectures and reasoning tools [6]. Fuenmayor and Benzmüller have formalised, using the ENoRME approach, some first ambitious ethical theory [9].

3 ENoRMEous Plans

Progress in Machine Learning (ML), in particular deep learning, has enabled impressive recent success stories in the development and deployment of IASs. ENoRME will target pressing challenges that these successes have generated. A major concern is that IASs that rely exclusively on non-symbolic technologies increasingly lack transparency, explainability, verifiability, and ethical behaviour – the essential requirements for “Trustworthy AI” [13].¹ A particular challenge concerns the development of mechanisms of ethical and legal control for future IASs,

¹We acknowledge the valuable efforts being made inside the deep learning research community towards the extraction of higher-level representations from modularised neural networks. However, there are still many open research questions. Regarding interpretability of outcomes (cf. “explainable AI”), such mechanisms are rather laborious and only useful in a kind of ‘forensic’ way. AI systems need explicit ‘on-line’ interpretability.

such as the ability to construct and reason with high-level conceptual representations of ethical and legal concepts. A convincing solution must fruitfully integrate non-symbolic with symbolic techniques, based on explicit knowledge representation and reasoning.

The success of deep learning is based on the availability of huge amounts of data, and relies on the impressive growth in hardware capabilities that allow for massively parallel data processing. The area of automated reasoning in expressive logics is lagging behind in this regard. A building block that is missing on the symbolic side is powerful automated reasoning technology for expressive non-classical logics and their combinations, as required for realising adequate forms of automated and semi-automated normative reasoning. The development of competitive automated reasoning technology for such ambitious logics typically requires substantial resource investment and expertise, which is typically not available to small research teams or small-to-medium sized enterprises (SMEs). It is clear that any strategy aimed at democratizing access to AI, in particular for SMEs and educational institutions, must facilitate their access to such critical hardware resources. Developing, sharing, and standardising such technology to stimulate research and deployment is one important mission of ENoRME.

The ENoRME vision has been specifically geared towards simultaneously addressing these two concerns: developing mechanisms of ethical and legal control, and providing the necessary reasoning infrastructure for that reasoning. In addition to boosting knowledge transfer of research outcomes with the scientific community, industry, and the public sector, ENoRME will address the real need of SMEs and educational institutions for experimenting with AI *at scale*, with little risk and low cost. A primary focus thereby is on the provision of technology for automating expressive normative reasoning. This is beneficial for AI research in a wide sense, not only for the legal and ethical governing of IASs. ENoRME will develop, as its core deliverable, an integrated *cloud-based, on demand, Universal Reasoning Workbench (URW) with a specific focus on normative reasoning*. This URW will support modelling and automated experimentation with explicit theories for use in ethical governors within IASs. The URW will also find application in other areas, e.g., rational argumentation, natural language processing, computational metaphysics, etc. Moreover, all of ENoRME's solutions will be applicable both independently and in combination with alternative means based on ML and other non-symbolic approaches. The overall objective is to contribute a missing *explicit reasoning* component in the development of secure, explainable, and trustworthy IASs.

At the same time as providing pragmatic solutions, ENoRME will engage in theoretical and experimental research in explicit ethical governing mechanisms for IASs. In this sense ENoRME will become its own first customer. The idea is to create an immediate feedback loop between (i) the URW, (ii) the ethical governing architectures, and (iii) their experimental application in selected case studies with our industrial partners. This feedback loop will then particularly inform our work on the URW.

The specific deliverables of ENoRME are as follows: • A cloud-based, on demand, URW, including an associated experimentation platform; • Standardisation and benchmarking in normative reasoning; • Theory and implementation of novel logic combinations; • Examples of mechanised/automated ethical and legal theories; • Specialist ATP and model finding technology for expressive normative reasoning; • An agent-based simulation environment enabling experiments with ethical and legal theories; • Case studies conducted within this agent-based simulation environment to assess different ethical and legal theories; • Exemplary implementation, experimentation, and assessment of ethical governor architectures equipped with ethical and legal theories; • Conferences, workshops, and tutorials on the topics of ENoRME; • Lecture courses, seminars, eLearning resources, PhD and student projects, all on the topic of ENoRME; • Outreach: public debates, media appearances, science slams, website, Facebook, Twitter, etc.

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